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Circular Prototyping: Detect - Conduct - Disrupt

T. Hegli, K. Edwards & L.E. Broström
The Oslo School of Architecture and Design (AHO), Oslo, Norway

J.K. Saunders
Asplan Viak, Sandvika, Norway

A.J. Petersen
The Norwegian University of Life Sciences, (NMBU), Ås, Norway

ABSTRACT: This paper describes the work of the Circular Prototyping MA studio at Oslo School of Architecture and Design. It seeks to determine the value of fluid transitions between theory, pedagogy and practice within the emerging realm of architecture with planetary boundaries and circularity as an applied framework. The paper presents three case studies of realized design prototypes as demonstrators of policy frameworks applied at the academic level as a foundation for design development. In parallel, resistive design parameters such as GWP and resource economies provide conditions for advancement in the pedagogy and practice of circular building practices. Finally, the discussion relates the value of course learnings to practice and how policy making can benefit from a proving ground hosted by academic institutions. The paper discloses opportunities in the broader adoption of prototyping as a method to fully realize the potential of the architect as a polymath.

1 INTRODUCTION

As regions grapple with consequences of multiple ecological tipping points, the architect must adapt to catalyse necessary change in practice to halt environmental degradation and climate gas emissions. A hypothesis therefore is that the architect of the Anthropocene has two equally important roles: to create protection from climate for human habitation, and to protect climate from human activity. The discord between the ability to comprehend and therefore assertively confront wicked problems such as human-induced climate change, and biodiversity collapse has become the defining challenge of contemporary architecture.

The planetary boundaries framework provides a science-based understanding of environmental limits within which humanity can safely operate (Richardson et al. 2023). Ratified intergovernmental agreements, such as the *Kunming-Montreal Global Biodiversity Framework* (UNEP 2022), emphasize nature protection and restoration, tying these goals to actionable policies. In the construction industry, decarbonizing the energy mix and implementing regulatory measures for energy consumption have been effective strategies. However, with growing concerns regarding resource scarcity and environmental impacts from raw material extraction, UNEP's report *Building Materials and the Climate: Constructing a New Future* (UNEP 2023) calls for a shift toward circular economies in building material production. A shift in focus by policy initiatives e.g. *EU Policy Whole Life Carbon Roadmap* (World Green Building Council 2022), calls for decarbonization of operational energy and embodied emissions as representing an essential step in decarbonization of the sector. (Röck et al. 2020).

In Norway, the inter-municipal body of FutureBuilt (Resch et al. 2021) facilitates pioneering project initiatives in the construction industry to reach targets for these global objectives by creating guidelines for sustainable development. These guidelines integrate qualitative and quantitative criteria, spanning diverse typologies and programs, as complementary to commercial building certifications. As-built

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results from pioneering pilots, alongside a continuous update from the research environment within the respective fields, establish a recurring feedback loop between academia and practice, further influencing policy-making with built cases as proving ground. Over the past decade, state-of-the-art projects have demonstrated measurable Whole Life Carbon (WLC) reductions, highlighting the necessity of near-term decarbonization and scalable circular practices to address resource depletion, waste, and pollution. However, accelerating circular practices comes with risks, necessitating faster, more methodological approaches to avoid missteps and burden-shifting.

This paper positions the Circular Prototyping MA studio as a critical testing ground for advancing circular building practices in collaboration with public and private stakeholders. With a solid professional foundation in projects aligned with FutureBuilt frameworks, the Circular Prototyping studio teaching environment bridges theoretical and practical gaps, aiming to influence regulatory requirements, drive innovation, disrupt value chains, and foster systemic change, positioning future architects to make a difference.

The Circular Prototyping studio adopts a design-build pedagogy, applying FutureBuilt criteria for circularity (Nordby et.al. 2023) and decarbonization strategies (Resch et.al. 2021) as foundational understanding. Structured across three phases - *Detect, Conduct*, and *Disrupt* - the architectural assignment encompasses a designated circular material strategy, opening for prototyping of circular building practices as integral to design-build practice. This generates findings that establish a recursive model for iterative development across studios and projects. The approach enhances knowledge transfer from theoretical research to real-world applications, creating a dynamic feedback loop between *designing* and *building*, positioning architectural practice as driving force in a continuous evolution of circular building practices. The architectural examples presented in this paper encompass prototyping of three circular material economy strategies: *Repurposing, Recycling*, and *Regenerative*.

2 DETECT

2.1 *Beyond system boundaries*

Life cycle assessment (LCA) quantifies energy use, resource depletion, greenhouse gas emissions, and other environmental indicators across the entire lifecycle of a product or building. Many nations have standardized LCA methodology to calculate greenhouse gas emissions resulting from building projects. However, this framework can be problematic when it comes to projected end-of-life scenarios for the material resources embedded in modern buildings. Rigamonti and Mancini (2021) point out that LCA methodology is not fully capable of measuring circularity for two main reasons. First, LCA models take a linear economy as the frame of reference; products are analysed from cradle to grave, ending in disposal of the product. Even with the eventual incorporation of reuse and recovery, modelling of open recycling loops remains inconsistent when accounting for multiple material uses with changing material qualities. Second, the impact indicators in LCA are calculated using characterization methods based on consumption, meaning that resources dissipated in anthropogenic deposits are treated as unavailable. Their study argues that LCA should be supplemented with separate indicators for circularity to get a more complete view of the studied system.

To promote a circular material economy alongside decarbonization, greater emphasis must be placed on inherent qualities and architectural design strategies that enable materials to remain in circulation beyond their initial service life, after a building is dismantled. Today, these capacities fall outside of LCA system boundaries, giving little incentive for systemic change in the construction sector.

2.2 *Decarbonization strategies*

LCA analyses often highlight a tension between emissions from energy use (operational carbon) and emissions from material production (embodied carbon) when it comes to utilizing existing buildings, components, materials or techniques with lower energy performance than today's industry standards. The alignment of these measures is an evident way to promote the environmental value of circular building practices, actualizing the utilization of existing resources as the most efficient strategy to reach climate neutrality by 2050. Regulatory standards and comfort criteria impose high demands on energy

performance, often leading to prematurely discarding functional materials like mechanical equipment and double-glazed windows. In response, architecture is increasingly embracing low-tech concepts displayed in traditional geo-based and bio-based construction with minimal processing, such as clay, hemp, and timber. These materials leverage inherent properties to provide comfort and well-being with fewer technical layers, aligning with regenerative design principles. While reuse, repurpose and low-tech solutions represented in regenerative building principals reduce initial GHG emissions, their use-stage performance requires attention. This approach entails need for post-occupancy measures to determine if lifecycle emissions targets are reliable (Hossain et al., 2021). Scaling regenerative materials requires addressing indoor climate and energy use to balance low embodied carbon with occupant comfort and resulting operational carbon in a lifecycle perspective.

3 CONDUCT

3.1 *Tethering to practice*

Considered by many schools as a vital component in modern architectural education, design-build pedagogy aims to prepare students for real-world challenges through active learning and engagement with material construction. Critiques of the pedagogic method describe dissonance in translation and application between practitioners and resistance to theoretical inference as typical (Erdman et al. 2002). As a potential answer to this critique, the transition of the design-build pedagogy to a prototyping approach adds strength to the entangling of design-discovery process and theory-construction process as creative and reflective acts respectively (Gleister 2022). Further argument is added by strong tethering to industrial initiatives based on research.

3.2 *Emerging pedagogies*

The three phases in the Circular Prototyping pedagogy - *Detect, Conduct* and *Disrupt* – aim at bridging form-finding processes with circular building practices and non-standard materials to foster a prototyping approach that supports the transition from a linear to a circular economy, encouraging architectural exploration to drive industry shifts.

The *Detect* phase in the Circular Prototyping pedagogy structure introduces lifecycle methodology with the LCA framework to understand what is included and what lies beyond, what can be quantified, and by which methods. To fully harness the architectural potential of lifecycle methodology, the *Detect* phase incorporates building physics, focusing on humidity transport and thermal performance (e.g., U-values) to assess capacities and mitigate potential risks when exploring unconventional assemblies and construction techniques. Embedding this knowledge as a core element of resource and reference studies makes it a pivotal aspect of the initial investigations.

The *Conduct* phase introduces the program and context, encompassing site visits, local climate analysis, and user requirements. This phase is rigged as an informal design competition, where a jury of external stakeholders selects one concept - or a fusion of several - based on the submissions. The contributions form a library of innovative architectural ideas, serving as a platform for the studio to develop a comprehensive construction manual. Concluding the theoretical design process, this phase adopts a holistic approach with an embedded prototyping exercise.

The Disrupt phase involves collaborative hands-on construction with initiative takers and industrial partners. This phase combines prototyping experience, quantitative results, and projections to form reflective reporting, detecting further needs to advance resource efficiency, techniques, and performance. Subsequently, three Circular Prototyping projects are presented.

4 DISRUPT

4.1 *Repurposing*

LCA analyses often highlight a tension between emission from energy use and emissions from material production when it comes to existing building components. Focusing solely on energy efficiency is a relic of a too-narrow focus on certain environmental impacts at the cost

of others, failing to consider resource use and conservation. Energy policies centred on reduction in demand through performance upgrades have created a glut of replaced windows with inadequate insulation values, typically either landfill or hazardous waste disposal awaits fully functional yet defunct units. Meanwhile, the need to retain resources in use at a high level and regulatory adjustments to recognize the value of repurposed existing materials are underscored by scarcity in glass-grade sand and staggering GHG emissions from production of new performance glazing. In 2022, a revision to Norwegian building code mandated a resource audit of reuseable materials before demolition (Resirqel & Sintef Byggforsk 2023). Challenges remain, however, including limited storage for salvaged materials and regulatory restrictions on reuse. For architects, designing from resource archives and incorporating design for disassembly (DfD) adds complexity, necessitating prototypes and testing.

In 2020, the Circular Prototyping studio examined the consequences of material repurposing principles on construction and ecological sustainability. The project involved designing and building a community greenhouse in Vardø together with the local cultivation initiative DyrkVaranger. Due to its high latitude and isolated coastal location, the region lacks forest industries and has few locally produced building materials, resulting in added costs from long-distance transport both for materials in and waste out. Consequently, the community has a tradition of storing usable materials in vacant private lots and buildings, motivated by a salvaging mindset. Tapping into his existing culture was therefore ideal for an investigation of how a circular material economy might function at a local scale.

In Vardø, engaged citizens collected building materials and registered properties. This effort provided timber for structural use, flooring and partitions, and 400 double- and triple-glazed windows of varying standards. With this resource archive transferred to Oslo, the studio categorised windows by precise measurements and identified three workflows in the reconstruction process: separation of frames and glazing, fabrication of new frames and assembly of frames and glazing units. Traditional techniques using linseed putty intended to make future replacement or repurposing possible. Additional expertise was brought in to identify hazardous compounds and ensure safe handling, of de-framed units. Upon professional examination the number of reusable windows was reduced by one-third.

The greenhouse design brief called for a key-less structure to house educational programs and cultural events alongside the arctic greenhouse experiment. Design development included studies of climate conditions at this latitude. The modelling of solar gain determined proportions and orientation of glass facades. Five distinct designs created a library of repurposing strategies for the joint studio effort to subsequently develop a construction manual for the chosen scheme.

The Greenhouse has proven successful across a range of programs and has become an iconic symbol for a community in need of social spaces to gather, share, learn, and grow. The loose-fit spatial plan is active throughout the year, with the highlight in September when the children's school harvests arctic cucumbers.



Figure 1. The Community Greenhouse under construction, in situ and in use.

The studio's concluding assessments, indicate that repurposing strategies in this location have potential to reduce environmental impacts by 80% when compared with import of new windows and export as waste as end-of-life scenario. The conclusions around the practical implications of re-framing were, however, less optimistic. The reframing process entailed safety and quality measures not easily met without expertise, and traditional assembly techniques despite re-use opportunities proved prohibitively time consuming. An eventual upscaling would require a systematic documentation of window products and a professional entity to potentially restore or replace frames where needed, ensuring compliance with toxicology regulations in handling, as well as secure technical adaptations allowing for new uses. From the perspective of design, the stripped-down glazing units were fragile, and adequate adaptation to breakage and replacement format was not provided in the design phase. Additionally, the co-dependent relationship of the load-bearing structure and glazing formats led to inefficient use of structural capacities and irrational spacing. Returning to the point of departure; realization of the community Greenhouse, served to broaden the perspectives of the citizens of Vardø, showcasing the environmental benefits of repurposing as a valuable product of their culture of material salvage.

4.2 *Recycling*

Plastics, accounting for at least 85 percent of total marine waste, are the largest, most harmful, and most persistent fraction of marine litter (UNEP 2021). Norway's commercial fishing sector alone generates 4000 tons of plastic waste annually from fishing gear and ropes. Just 26% of this waste is recycled, and only a small fraction of that is recycled within the region (Deshpande et al. 2020). With the European Commission action plan on Reducing Marine Litter (EC 2018), initiatives are emerging to close the loop and ensure sustainable resource management using circular strategies. One example is the Oceanize company, which produces granulate from discarded rope, avoiding negative environmental impacts from landfilling and incineration. The product contains a mix of polymers, resulting in a loss of properties when compared to virgin plastics. This means that new rope products cannot be made from the granulate meaning it is regarded as down-cycled media. There remains potential for use in the building industry as replacement for virgin plastic products with less stringent requirements, but few examples are found today, meaning more data on opportunities and consequences is needed.

In the spring of 2023, the environmental collective Nordic Ocean Watch (NOW) initiated a collaboration with the Circular Prototyping studio to realize a temporary pavilion to house an educational program on awareness of marine plastic awareness. The project was funded by Oslo Municipality and other public actors, and the pavilion given a central location next to the Oslo Opera house. With the pavilion project as a testing ground, a value chain was constructed where Oceanize delivered granulate to the injection moulding company Plan-Mek, who produced fixed-format panels for use as weather proofing using a borrowed injection mould from Hofseth Aqua, a Norwegian fishing industry actor. This recycled plastic panel provided a circular starting point for the pavilion's design development. Criteria for the design task further entailed detailing for disassembly to enable relocation of the pavilion and securing the plastic products return to Oceanize when the pavilion is replaced by a planned permanent centre. Alongside repurposed structural timber and rented floating rig, this collaboration provided a solution to how discarded plastics could find refuge as building elements providing shelter to those participating in educational activity on ocean waste.

While the programming successfully facilitates discussions on why plastic goes astray and explores solutions involving the community, authorities, and the business sector, the architecture speaks on the necessity to address the problem with closed-loop value chains at the industrial level. Findings from the circular resource experiment show that despite the obvious environmental benefits of preventing GHG emissions, and landfilling through recycling of discarded rope, the risk of leaching was not possible to quantify or address without further lab testing. As a result, a research initiative by Oceanize will verify the panels after a certain



Figure 2. The pavilion in production, in situ and in use.

period of exposure to the elements. Results are pending. To get a grip on the wicked problem of plastics in the context of the building industry, the priority should be to diminish use of plastics at large, as reflected in line in the FutureBuilt guidelines (Hagen & Smith, 2022).

This case underscores both the importance and difficulty of avoiding burden-shifting: addressing one environmental issue while creating another. Other implications for a safe transition to circular material practices typically unaddressed by assessments are how to account for knowledge gaps – for example, chemicals from manufacturing processes that might be considered safe in the context of their original use but, can be harmful when repurposed or recycled to a different state.

4.3 *Regenerative*

With the urgent need to mitigate climate change, reducing GHG emissions at initial stages of construction must take priority over long-term predictions with significant uncertainties. Low-carbon and biogenic materials are subject to increased attention, emphasizing the necessity of adapting traditional techniques into applicable standards for contemporary industry (Beim et al. 2023). Another emerging field is the terminology used to categorize and specify natural materials based on techniques, whether as elements or in-situ applications, and sourcing, such as local or manufactured. Circular Prototyping defines regenerative materials as those that minimize environmental impact during the product stage and actively restore and enhance ecosystems as part of an optimal end-of-life scenario. Clay and clay-based building techniques, as a subject field for Circular Prototyping, exemplify this category.

Clay, an abundant biogenic material in the Norwegian landscape, has served an essential role in traditional building methods such as dry-stone masonry, cordwood construction, and plaster finishing. When mixed with biogenic fibre, soil or sand, clay acts as a natural binder, forming a cohesive bio-composite upon drying. The combination of lightweight organic fillers - such as straw, wood chips, or hemp, results in a light-clay mixture gaining additional strength and insulation capacity. Dependent on the amount and type of fibre in the mix, assembly and construction techniques call for theoretical and practical investigations to determine performance and detailing according to the local climate conditions, site, and intended use (Berge et.al.2009).

In the spring of 2023, the Circular Prototyping studio teamed up with Sperlebakken, a small farm in Luster, initiating cultural activities and regenerative cultivation. The assignment was to realize a small lodge as a first step to establishing an artist-in-residence program. As the intended use of the lodge was primarily to give artists in residence a place to retreat, meditate, and sleep in between creative productions, attention to daylight and view played a vital role in Sperlebakken choosing their preferred architectural scheme.

Designed as an off-grid retreat with natural light and a wood-fired stove for heating, the vernacular construction technique of in-situ light-clay (woodchip clay) as infill in a timber stud frame, was found suit the architectural intent. Preparation for the design phase included an in-depth investigation of contemporary geo-based construction across Europe to provide knowledge on mixture recipes, inherent material capacities and thermal performance. Architectural details to protect exposed clay surfaces became critical and parallel investigations focussed on sourcing, GWP, and building physics to generate a complete picture of the methodology required to introduce light-clay techniques into a Norwegian context of culture, climate and industry.

The construction phase provided valuable insight into the necessities of constructing with these techniques. With the spectacular site in a deep-lying fjord valley, local climate conditions with high relative humidity elongated drying times creating issues with the clay-based plaster. Consequently, blending and application of the woodchip clay media depended upon many hands. With the rather poor u-value, reflections on the program balance with operational energy use and comfort standards, triggers discussions on behavioural changes as a central caveat for application of these techniques going forward. While local applications requiring sensitivity to seasons and weather changes suggest an act of passion rather than industry, numerous discoveries were made to suggest that industrial scale-up could be feasible with regional manufacturing reducing transport emissions, yet complicated without prefabrication in a controlled environment. Furthermore, concerns relating to degeneration of ecosystems and land depletion as result of extraction processes should be considered as an unwanted byproduct of a potential industrial scale-up.

5 DISCUSSION

With continually updated industry-guidelines providing conditional methodology to the Circular Prototyping pedagogy, student projects function as pilots in a feedback loop. The



Figure 3. The Clay wood chip, hut under construction and in situ.

pedagogical framework of Detect-Conduct-Disrupt allows for rapid stress-testing, attacking real-world challenges. By conducting full-scale experiments in a short time frame, gaps between theory and practice quickly become clear, building experience proving crucial for component reuse architecture with inherent challenges of adapting design to the challenge of singular products. Reporting on the realized architectural project typically exposes a lack in the systems, techniques and craftsmanship required for success, thus acting as an early demonstrator of critical methodologies to adopt and scale boundary-based budgets in new building standards.

The circular prototype constitutes the design-build project, and the architectural design drawings and reported findings, consequently through this feedback loop, pilots provide incremental advancement to both the sector as a whole and architectural practice specifically. FutureBuilt acts as a two-way conduit, translating research on the impacts of the built environment, land use, biodiversity, and circularity into actionable guidelines for design practices and the building industry. Presentation of studio processes and critical findings at open seminars directed toward architectural practices and professional actors hosted by FutureBuilt, academic dissemination potentially offers industrial disruption.

The pedagogical framework also acts as an advancement to aesthetic evaluation, to where environmental relevance earns credibility and focus. By learning to identify environmental considerations and trade-offs, students develop another lens through which to view potential designs (Detect). Pedagogic facilitation ensures lifecycle considerations become an integral part of design decision making (Conduct), and assumptions and hypotheses are challenged during real construction (Disrupt). For architects, designing from resource archives and incorporating design for disassembly (DfD) is complex, making testing through full scale prototypes an increasing necessity. Students learn to look beyond surface appearance, carrying this knowledge and attitude with them throughout their career serves to strengthen architectural practice at the advent of the Anthropocene.

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